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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

3093404356

MATHEMATICS (SYLLABUS D) **4024/12**

Paper 1 **October/November 2011**

2 hours

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.

ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 80.

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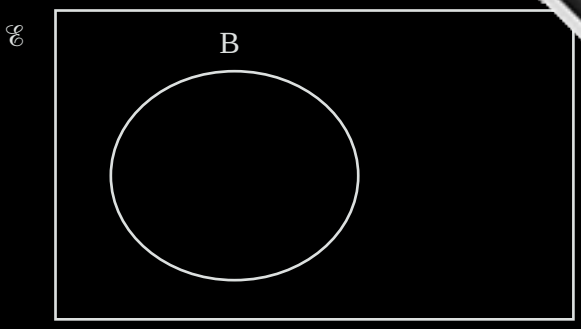
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[2]

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11 By writing each number correct to two significant figures, estimate, correct to one significant figure, the value of

$$\sqrt{110.94 - 0.2034 \times 368.62} \ .$$

Answer ..... [2]

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12 The length of a rectangle is 8 cm.
It is increased by 150%.

Calculate the new length.

Answer ..... cm [2]

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x	3	4	q	n
y	20	p	5	m

(a) Find p.

Answer $p =$ ..... [1]

(b) Find q.

Answer $q =$ ..... [1]

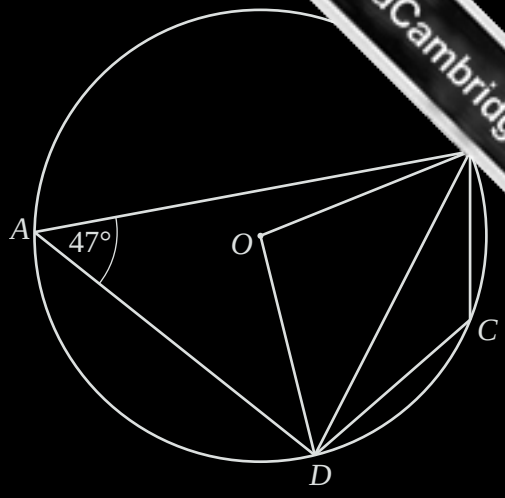
(c) Express m in terms of n.

Answer $m =$ ..... [1]

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the circle, centre O.

$B\hat{A}D = 47°$.



Find

(a) $B\hat{O}D$,

Answer $B\hat{O}D =$ ..... [1]

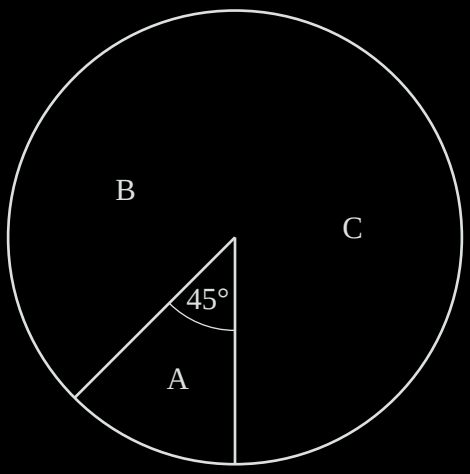
(b) $B\hat{C}D$,

Answer $B\hat{C}D =$ ..... [1]

(c) $O\hat{B}D$.

Answer $O\hat{B}D =$ ..... [1]

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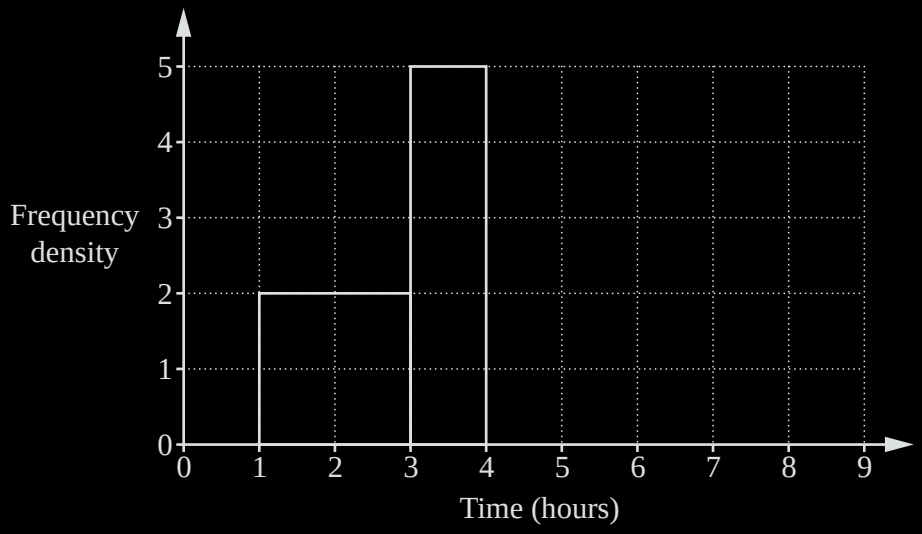
(b) Expressing your answer in its lowest terms, find the fraction of people in the survey who liked C best.

Answer ..... [1]

(c) Given that 30 people liked A best, calculate the number of people in the survey.

Answer ..... [1]

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(a) Find k.

Answer k = ... [1]

(b) Complete the histogram. [2]

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17 The length of a side of a square is given as 57 mm, correct to the nearest millimetre.

(a) Write down the upper bound for the length of a side.

Answer ..... mm [1]

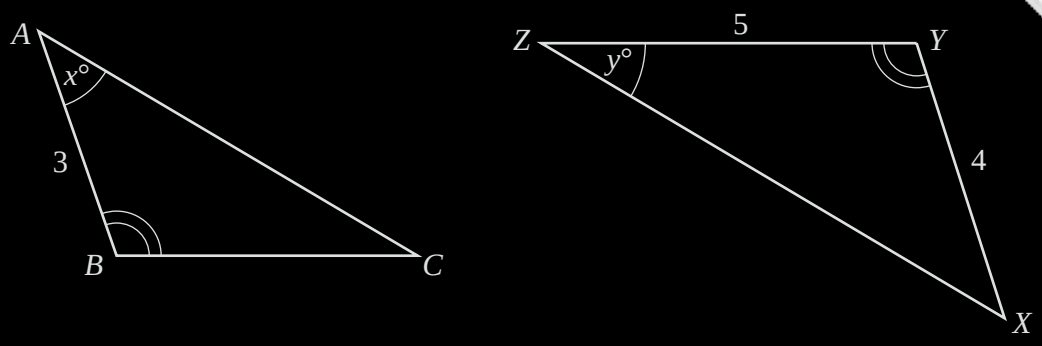
(b) Giving your answer in **centimetres**, calculate the upper bound for the perimeter of the square.

Answer ..... cm [2]

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The triangles ABC and XYZ are similar and $A\hat{B}C = X\hat{Y}Z$.

$B\hat{A}C = x°$, $Y\hat{Z}X = y°$ where $x \neq y$.
$AB = 3\,\text{cm}$, $XY = 4\,\text{cm}$ and $YZ = 5\,\text{cm}$.

(a) Express $A\hat{B}C$ in terms of x and y.

Answer $A\hat{B}C =$ ..... [1]

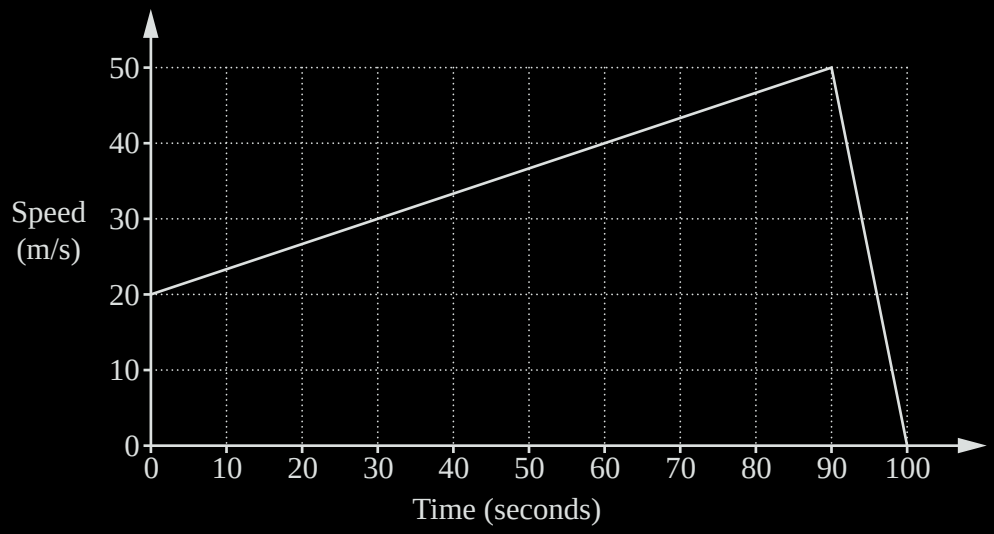
(b) Find BC.

Answer $BC =$ ..... cm [1]

(c) Write down the value of $\dfrac{\text{area of triangle } ABC}{\text{area of triangle } XYZ}$.

Answer ..... [1]

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The diagram is the speed-time graph of the last 100 seconds of a train's journey.

(a) Calculate the train's retardation during the last 10 seconds of the journey.

Answer ..... m/s^2 [1]

(b) Calculate the distance travelled in the 100 seconds.

Answer ..... m [2]

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Number of goals	0	1	2	3	4	5	6
Number of teams	16	6	6	6	4	0	2

Find

(a) the mode,

Answer ..... [1]

(b) the median,

Answer ..... [1]

(c) the mean.

Answer ..... [2]

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$x + y = 12\frac{1}{2}$,

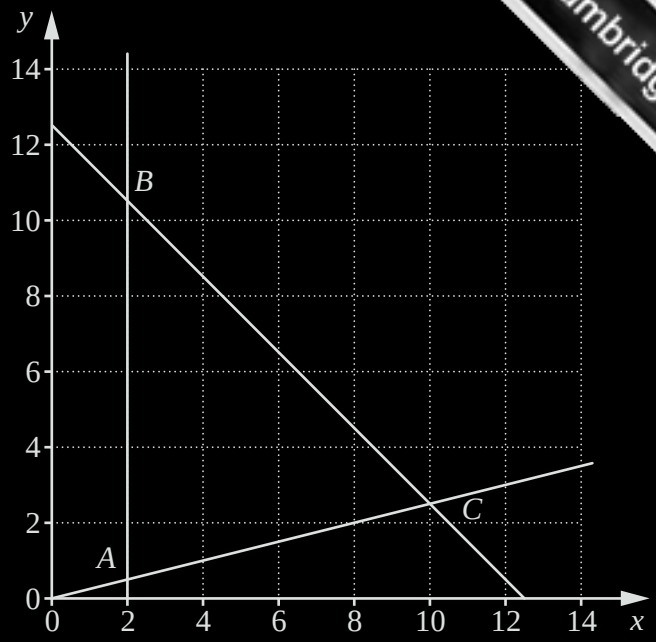
$y = \frac{x}{4}$ and

$x = 2$.

These graphs intersect to form triangle *ABC*.

The region **inside** triangle *ABC* is defined by three inequalities**.**

One of these is $y > \frac{x}{4}$.



(a) Write down the other two inequalities.

Answer .....

..... [2]

(b) P = {(*x*, *y*) : *x* and *y* are integers, (*x*, *y*) lies **inside** triangle *ABC*}
Q = {(7, *y*) : *y* is an integer}

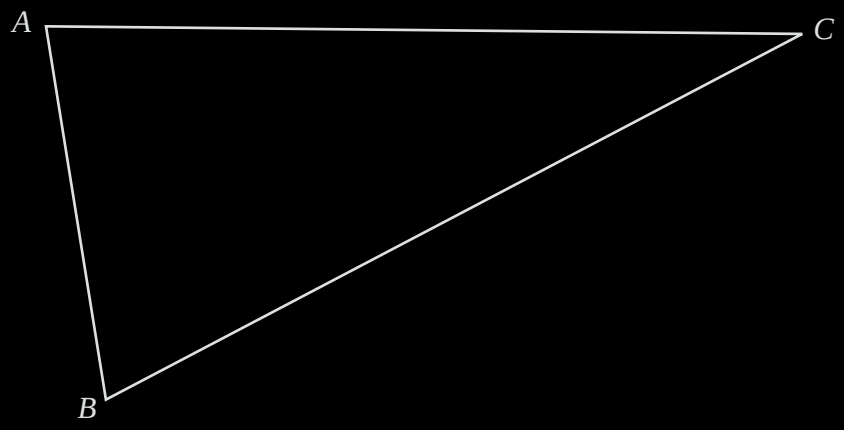
(i) Find the member of the set P that is closest to the point *C*.

Answer ..... [1]

(ii) Find $n(P \cap Q)$.

Answer ..... [1]

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(a) The point D is on the opposite side of AC to B.
$AD = 6$ cm and $CD = 8$ cm.

Construct triangle ADC. [1]

(b) On the diagram, construct the locus of points **inside** the quadrilateral $ABCD$ that are

(i) 2.5 cm from AC, [1]

(ii) equidistant from AB and BC. [1]

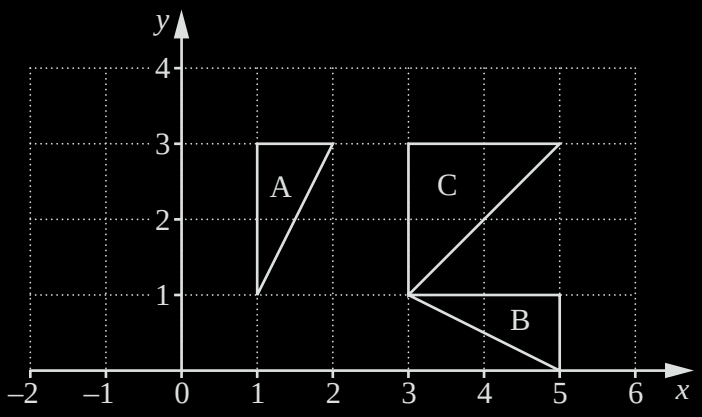
(c) The points P and Q are 2.5 cm from AC and equidistant from AB and BC.

Mark and label P and Q .

Measure PQ.

Answer $PQ =$ ..... cm [1]

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The diagram shows triangles A, B and C.

(a) Triangle A is mapped onto triangle B by an **anticlockwise** rotation.

(i) Write down the angle of rotation.

Answer ..... [1]

(ii) Find the coordinates of the centre of rotation.

Answer (..... , ...) [1]

(b) Triangle A is mapped onto triangle C by a stretch.

(i) Write down the scale factor.

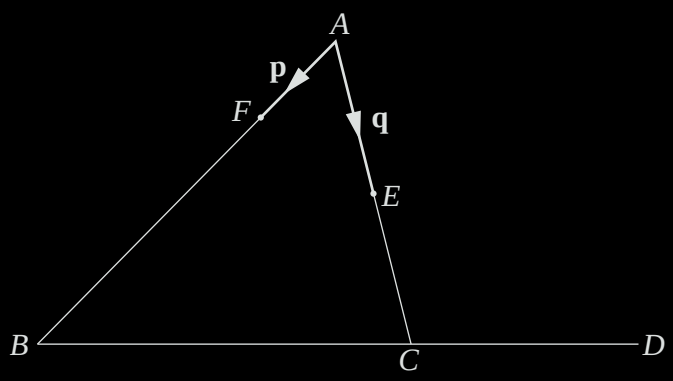
Answer ..... [1]

(ii) Write down the equation of the invariant line.

Answer ..... [1]

Question 28 is printed on the following page.

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In the diagram, F is the point on AB where $AF = \frac{1}{4} AB$.
E is the midpoint of AC.
$\overrightarrow{AF} = \mathbf{p}$ and $\overrightarrow{AE} = \mathbf{q}$.

(a) Express, in terms of **p** and **q**,

(i) $\overrightarrow{FE}$,

Answer ..... [1]

(ii) $\overrightarrow{BC}$.

Answer ..... [1]

(b) D is the point on BC produced such that $BD = kBC$.

(i) Express $\overrightarrow{FD}$ in terms of k, **p** and **q**.

Answer ..... [1]

(ii) Given that F, E and D are collinear, find the value of k.

Answer $k =$ ..... [2]

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